

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Weekly Progress Report — Week 2

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1 Introduction

After setting up the project and initial study of texture analysis. This week's work was on the two Local Binary Patterns, remaining texture-description techniques, a detailed examination of the AITEX dataset, and the conceptual design of the complete detection pipeline.

The work was still theoretical and exploratory with no implementation completed yet. In Week 3 coding will commence.

2 Literature Review: Texture Descriptors

2.1 Gray-Level Co-occurrence Matrix

The Gray-Level Co-occurrence Matrix (GLCM) describes texture by analyzing how frequently pairs of pixel intensity values occur at a specified distance and direction. From the resulting matrix, statistical features can be obtained to characterize properties such as texture uniformity, contrast, and regularity.

Multiple directions are important because woven fabrics have directional thread structures, and a defect may disturb the texture differently along different orientations.

2.2 Gabor Filtering

Gabor filtering analyzes texture by detecting repeating patterns at specific frequencies and orientations. This makes it particularly suitable for woven fabric, where the normal texture is formed by repetitive thread patterns.

Multiple Gabor filters with different frequencies and orientations can capture a more complete representation of the fabric texture. A significant change in the filter response may indicate a disruption caused by a defect.

3 Understanding the Detection Approach

With the three texture descriptors now studied, the team developed a conceptual understanding of how they will be combined into the proposed defect-detection pipeline:

- Establish the normal texture characteristics of each fabric type using defect-free images.
- Extract texture features from new images and compare them with the corresponding normal texture.
- Identify regions with significant texture differences as potential defects.
- Apply image-processing operations to remove noise and obtain coherent defect regions.
- Localize the detected defects and assign an appropriate confidence score.

This conceptual pipeline will serve as the basis for implementation starting in Week 3.

4 Dataset Characteristics

This week, the AITEX dataset was explored in more detail. The team Experimented on the various types of fabrics, type of defects and images. The connection of fabric images to its relationship with organization and to its corresponding reference images.

The pictures are not too small, and the aspect ratio is large, because of the resolution of the photos taken: In

what way the fabric was captured in manufacturing process. This will be taken into account to optimize the tradeoff between the number of computing nodes required and the cost of the implementation, the system should be preprocessed and implemented during the design while efficiently preserving relevant texture information.

5 Software and Implementation Plan

The first set of code will be broken down into various modules for:

- The method in which datasets are loaded and organized.
- Image preprocessing and the texture features will be extracted
- The normal-texture comparison and defect detection.
- Defect localisation and assessment

This is a modular design that will enable individual parts to be developed and tested independently as implementation goes on.

6 Work Completed This Week

- Learned about gray level co-occurrence matrix (GLCM) and its properties.
- Learned about Gabor filtering and its applicability for repetitive tasks.
- Developed an understanding of the whole concept texture-based defect-detection pipeline.
- Reviewed AITEX data set (fabric and defect types), The organization, management and retrieval of images.
- Created the first draft of the structure and implementation plan for the codebase.

7 Plan for Next Week

- Start pre-processing of the fabric images.
- Make initial training/testing split in data set.
- Start to implement selected texture descriptors.

End of Week 2 Report